Improving Health Literacy Through Accessible Design

Suhani Sengupta

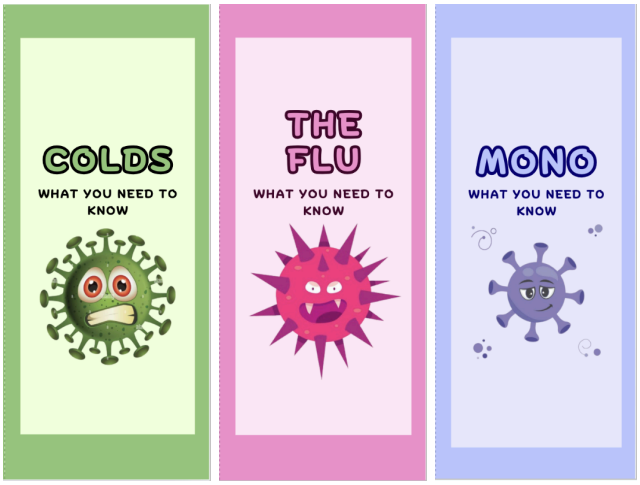
INTRODUCTION

Engineering is usually associated with building systems or devices, but in practice, it often comes down to solving everyday problems people deal with. For my service project, I focused on a communication issue in healthcare: many patients struggle to understand basic medical instructions. Discharge papers, medication labels, and follow-up care sheets are often written in technical language and assume prior knowledge. This can lead to missed medications, confusion, and improper care at home. I wanted to address this by making health information easier to read and actually usable.

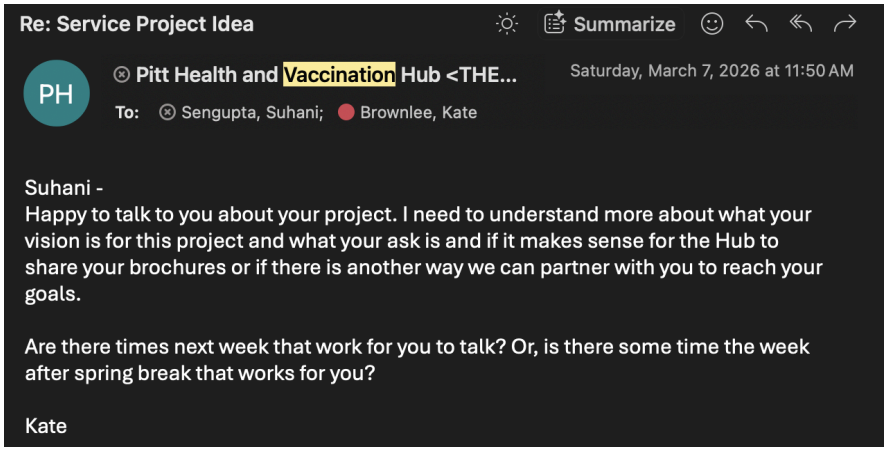
PROCESS

My original proposal was to create simple, clear health instruction guides that break down common conditions in a way that people can quickly understand. I designed brochures using Canva and structured them into sections like what the condition is, how it spreads, symptoms, what is normal versus when to seek help, what to do next, and prevention. I kept the language direct and avoided medical jargon, and I used visuals only when they made the information clearer. I also used bright, simple color schemes based on what I found in my research, since color and layout can affect whether someone actually pays attention to and reads the information. The goal was to make something that someone could pick up and understand right away, even if they were overwhelmed.

I created three brochures focused on common student health concerns: mono, the cold, and the flu (pictured below). I printed 40 copies of each brochure, for a total of 120, with the help of my academic advisor, Ms. Washko. These topics were chosen because they are both common and often misunderstood, especially when students are trying to decide whether their symptoms are serious or not.



To make sure this project was actually useful, I reached out to different organizations and talked to people about how patients receive information. I emailed multiple groups and had conversations with stakeholders, including the Pitt Vaccination Hub (shown below). These conversations made me realize the problem is not just the information itself, but also where and when people see it. That ended up changing my entire approach.


Re: Service Project Idea

Summarize

PH

Pitt Health and Vaccination Hub <THE...

Saturday, March 7, 2026 at 11:50 AM

To: Sengupta, Suhani; Brownlee, Kate

Suhani -
Happy to talk to you about your project. I need to understand more about what your vision is for this project and what your ask is and if it makes sense for the Hub to share your brochures or if there is another way we can partner with you to reach your goals.

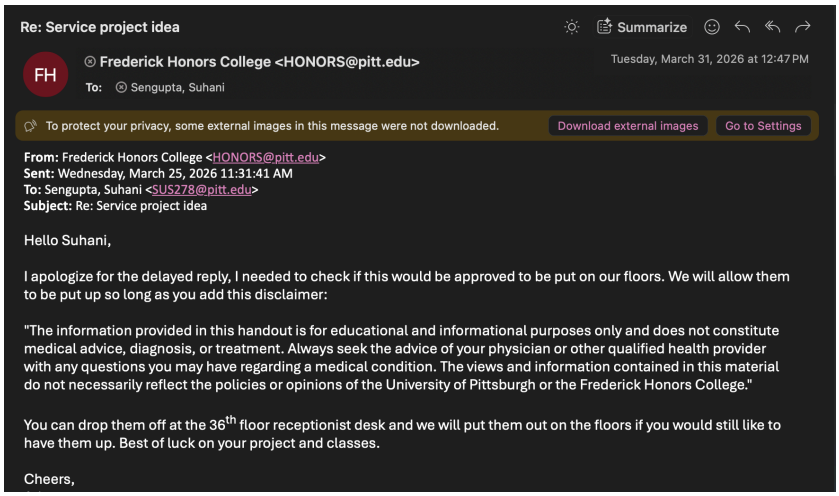
Are there times next week that work for you to talk? Or, is there some time the week after spring break that works for you?

Kate

Originally, I planned to place the brochures in clinical settings like the vaccination hub. But after meeting with them, they explained that most people there have already made a decision about their health. They are not really in a position where they are questioning or looking for new information. Because of that, placing brochures there would not be very effective.

Instead of sticking to my original plan, I adjusted based on their feedback. They suggested that it might be more useful to spread awareness of their resources through the brochures. So I changed the brochures to include a note with the vaccination hub's resources and address, so that people could find and use those services when they actually needed them.

After this, I changed my distribution location to the Pitt Honors College. This was a better fit because students there are more likely to be in the stage where they are unsure about symptoms or looking for information. I reached out to the Honors College (shown below), asked for permission, and was able to place the brochures there. This shift made the project more effective because it placed the information earlier in the decision process, where it could actually influence what someone does next.



Re: Service project idea

Summarize

FH

Frederick Honors College <HONORS@pitt.edu>

Tuesday, March 31, 2026 at 12:47 PM

To: Sengupta, Suhani

To protect your privacy, some external images in this message were not downloaded. Download external images Go to Settings

From: Frederick Honors College <HONORS@pitt.edu>
Sent: Wednesday, March 25, 2026 11:31:41 AM
To: Sengupta, Suhani <SUS278@pitt.edu>
Subject: Re: Service project idea

Hello Suhani,

I apologize for the delayed reply, I needed to check if this would be approved to be put on our floors. We will allow them to be put up so long as you add this disclaimer:

"The information provided in this handout is for educational and informational purposes only and does not constitute medical advice, diagnosis, or treatment. Always seek the advice of your physician or other qualified health provider with any questions you may have regarding a medical condition. The views and information contained in this material do not necessarily reflect the policies or opinions of the University of Pittsburgh or the Frederick Honors College."

You can drop them off at the 36th floor receptionist desk and we will put them out on the floors if you would still like to have them up. Best of luck on your project and classes.

Cheers,

EXECUTION

The execution of the project involved designing and refining the brochures, printing them, and placing them in a location where they would be seen and used. I also adapted the content based on what I learned from my conversations, especially by adding connections to existing healthcare resources instead of trying to replace them. The final brochures are not just informational, but also guide people toward actual services they can use (Mono example shown below).

HOW TO PREVENT MONO

REDUCE YOUR RISK!
- Do not share drinks, utensils, or toothbrushes
- Avoid kissing or close contact with someone who is sick
- Wash your hands regularly
- Cover your mouth when coughing or sneezing

KNOW THE REALITY
Mono is very common, especially in college settings. Many people are exposed to the virus at some point.

PROTECT OTHERS!
If you are sick:
- Avoid close contact when possible
- Do not share personal items
- Stay home and rest when symptoms are strong

THINK YOU MIGHT HAVE MONO?

If you are experiencing symptoms or feel unsure, it is important to seek medical advice.

Don't ignore your symptoms– getting checked early can help you recover faster and avoid complications.

Contact Student Health Services:
Phone: 412-383-1800
Email: shspitt@pitt.edu
Location: Nordenberg Hall – Wellness Center
119 University Place

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MONO

WHAT YOU NEED TO KNOW

WHAT DO I DO IF I HAVE MONO?

REST!
- Your body needs time to recover. Avoid overworking yourself, even if you feel a little better.

STAY HYDRATED!
- Drink plenty of water. Warm fluids can help with a sore throat.

MANAGE SYMPTOMS!
- Use over-the-counter pain relievers like ibuprofen or acetaminophen)
- Gargle salt water for throat pain
- Eat soft foods if swallowing hurts

AVOID INTENSE ACTIVITY!
- Mono can cause your spleen to swell. Avoid sports, especially contact sports, for several weeks to prevent injury.

LIMIT CONTACT!
- Try not to share drinks, food, or utensils to avoid spreading it.

IMPORTANT:
- Antibiotics do not treat mono because it is a viral infection.

COMMON SYMPTOMS

These are typical and expected:
- Extreme fatigue (feeling very tired all the time)
- Sore throat (often severe)
- Fever
- Swollen lymph nodes (especially in the neck)
- Headache
- Body aches
- Loss of appetite

WHATS AN "OKAY" REACTION?

These symptoms can feel intense but are usually manageable:
- Feeling tired for weeks
- Mild fever
- Sore throat that improves slowly

WHAT'S MORE SERIOUS? (GET MEDICAL HELP)
- Severe abdominal pain (could mean spleen swelling)
- Trouble breathing or swallowing
- Very high fever that doesn't go down
- Extreme weakness where you can't function
- Yellowing of skin or eyes

If something feels worse than expected, it's better to get checked.

WHAT IS MONO?

Infectious Mononucleosis is a viral infection that is common in teenagers and college students. It is often called the "kissing disease," but it can spread in other ways too.

WHAT CAUSES IT?

Mono is usually caused by the Epstein-Barr virus (EBV), which is a very common virus.

Once you get it, the virus stays in your body for life, but it usually becomes inactive after some time.

HOW DOES IT SPREAD?

Mono spreads through saliva. You can get it by:
- Kissing
- Sharing drinks, utensils, or food
- Using someone else's toothbrush
- Close contact like coughing or sneezing

You can spread mono even if you don't feel sick yet.

The impact of this project comes from both the design and the placement. The brochures simplify complex medical information into something that is structured and easy to follow. At the same time, placing them in a student-focused environment increases the chance that people will actually read and use them (brochures placed on Honors college floor shown below). Even small changes in how information is presented can make a difference in whether someone understands how to take care of themselves.

This project also showed me that engineering is not just about coming up with a solution, but about adjusting that solution based on real feedback. My original plan was not placed in the right context. Through interviews and conversations, I was able to change my approach and make it more effective. That process of testing, learning, and improving is what made the project meaningful.

CONCLUSION

Overall, this project reflects how engineering can support public health in a practical way. By focusing on clarity and usability, I was able to create something that addresses a common problem in a simple and effective way. This project showed me that even small changes in how information is presented can have a real impact on how people understand and manage their health.